

Regular Articles

Purification technique with a hertz-linewidth laser for fiber-based optical frequency coherence dissemination

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ABSTRACT

We propose a purification relay station for optical frequency dissemination over long-distance communication fiber links particularly in complex environments. This technique addresses the challenge of phase noise suppression limitation due to dissemination delay in laser traversing fiber links. Utilizing a cavity-stabilized hertz-linewidth laser, we synchronized the phase noise with that of the signal received from the previous link within a 10 Hz bandwidth. Beyond this range, the phase noise followed the hertz-linewidth laser, decreasing from 20 rad to 6 rad. Furthermore, to ensure continuous coherence and stability of the disseminated laser, the data processing based on FPGA was employed to eliminate phase cycle slips in the previously disseminated optical frequency. We applied this technique to the optical frequency disseminated along with a 106 km communication fiber link. After optical frequency purification, the fractional frequency instability at a 1 s averaging time is reduced by a factor of three. Additionally, with the elimination of the phase cycle slips, the fractional frequency instability of purified optical frequency scales as $\frac{1}{\sqrt{t}}$ which matches with a signal dominated by phase-modulated white noise. At a 10,000 s averaging time, the purified optical frequency achieves a stability of 8.43×10^{-19} , representing a reduction of two orders of magnitude compared to that of the previously disseminated optical frequency. This work establishes the foundation for the development of a nationwide optical frequency dissemination network involving long-distance fiber links in complex environments.

1. Introduction

Researchers at numerous international and domestic institutions have developed optical atomic clocks with fractional frequency uncertainties of 10^{-17} level or greater [1–3]. Furthermore, the 27th International Conference on Metrology (CGPM) passed a resolution to redefine the second in the future. Time scales based on one or more optical frequency standards will be considered in the new second definition, which is expected to be adopted at the 29th meeting of the CGPM in 2030 [4]. To practically apply ultrahigh-precision optical frequency standards, an ultrahigh-precision time–frequency dissemination technique is needed. The extensively studied fiber-optic frequency dissemination technique has proven to be instrumental in disseminating optical frequency over distances exceeding 1,000 km, achieving frequency dissemination instability better than 10^{-19} [5–7]. This technique is crucial for accurate comparisons of optical clocks at remote locations [8]

and facilitates their application in metrology [9], geodesy [9,10], fundamental science [11], and dark matter research [12].

Effective phase noise cancellation (PNC) of fiber links is paramount for achieving high-precision optical frequency dissemination through long-distance fiber links. In 2010, Olivier Lopez et al. demonstrated a cascaded optical link for ultrastable frequency dissemination composed of two compensated 150 km links and a repeater station, with the disseminated optical frequency having an instability of 5×10^{-20} after 20 h [13]. The phase-locked loop bandwidth can be extended through the repeater station to improve the phase noise suppression ratio. However, as the number of repeater stations increases, a servo bump appears at the control bandwidth, which affects the locking of subsequent links [14]. In 2020, L. Hu et al. proposed a passive PNC technique for optical frequency transfer via an open-loop compensation strategy. This technique aims to suppress phase noise introduced by servo bumps, but the suppression ratio of the phase noise is lower than that of the

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conventional multiple-access technique when the access point is close to the end of the fiber link [15–17].

Continuous optical frequency dissemination is needed for ultraprecise frequency comparisons with uncertainties below 10^{-18} level. The phase noise of the fiber links in long-distance optical frequency dissemination is not strictly stationary; therefore, interruptions inevitably occur in the form of cycle slips, which reduce the long-term frequency instability in the dissemination. This deterioration is related to the magnitude and number of the phase cycle slips and their locations in the data [18]. To achieve frequency transfer through the Global Navigation Satellite System (GNSS) and two-way satellite time and frequency transfers, several evaluation and correction techniques have been developed to address the missing data caused by interruptions [19–22]. Mads Tønnes et al. addressed these challenges for optical frequency dissemination via fiber links and explored three different approaches for addressing missing data in a 1,410-km-long optical link. Their findings suggest the advantages of replacing missing data with simulated noise that exhibits statistical behaviour similar to that of the original signal [23,24]. However, these methods involve post-collection data processing, and for certain applications requiring real-time measurements, the effectiveness in mitigating the impact of cycle slips is not enhanced.

In this paper, we propose an optical frequency purification technique that enables the phase of a local cavity-stabilized narrow linewidth laser to be locked to the disseminated light phase with a bandwidth of 10 Hz. The proposed technique aims to reduce the phase jitter caused by the servo bump at the control bandwidth while improving the phase noise suppression ratio and eliminating the cycle slips caused by external interference. Moreover, the proposed optical frequency purification technique aims to maintain the coherence of the fiber links and achieve excellent short-term and long-term frequency dissemination instability, thereby improving the overall reliability and precision of the disseminated optical frequency. This approach can be combined with the cascade dissemination technique to achieve better optical frequency instability in long-distance dissemination systems.

We apply this technique to the optical frequency disseminated over a 106 km communication fiber link. We investigate the phase noise and instability of the purified and disseminated optical frequency and compare the results. The phase noise of the purified optical frequency is consistent with that of the disseminated optical frequency within a bandwidth of 10 Hz and follows the phase noise of the narrow linewidth laser beyond 10 Hz (much lower than the phase noise of the disseminated optical frequency). The instabilities of the disseminated and purified optical frequency are 1.57×10^{-14} and 4.50×10^{-15} , respectively, at an averaging time of 1 s, reflecting the significant enhancement in the phase noise suppression ratio with the proposed technique. Furthermore, the phase cycle slips in the disseminated optical frequency are eliminated in the purified optical frequency, leading to an instability of 6.15×10^{-19} at a 10,000 s averaging time. This instability is two orders of magnitude greater than that of the disseminated optical frequency. This result validates the efficacy of the purification technology and guarantees the high precision of optical frequency dissemination.

2. Optical frequency purification

The optical frequency purification approach produces a smoothed optical frequency from a local narrow linewidth laser locked to the incoming disseminated optical frequency. Moreover, this approach can reduce the noise at high frequencies and repair discontinuous phase cycle slips in the incoming signal. The local narrow linewidth laser is an ultrastable laser (wavelength 1550.12 nm, linewidth 2 Hz) stabilized in an optical cavity (cavity length 50 mm, finesse $\sim 450,000$, coupling efficiency more than 50 %) using the Pound-Drever-Hall (PDH) method [25]. The output frequency of the local narrow linewidth laser is strongly affected by the ambient temperature, the relative frequency drift between the laser and the disseminated light is nonlinear and large (approximately 2 MHz/day), and traditional phase-locked loops are

difficult to implement. Therefore, a two-level control system is needed to address the large nonlinear relative frequency drift. In the first level, the relative frequency drift is eliminated through an open-loop compensation approach. Then, in the second level, the residual frequency and phase differences that need to be corrected to ensure precise coherence between the purified light and the light source are addressed through a closed-loop compensation strategy. Fig. 1 shows a schematic diagram of the optical frequency purification system. The incoming signal from the remote end of the previous link is split into two parts at a ratio of 90:10 by a single-mode coupler 1 (SMC1). The 10 % portion and 10 % of the local narrow linewidth laser are sent to SMC2 at a ratio of 90:10 for heterodyne detection, and the output signal of photoelectric detector 1 (PD1) is sent through the frequency compensation module and then fed back into acousto-optic modulator 1 (AOM1). This represents the first level of the control system, in which the relative frequency drift between the local ultrastable laser and the incoming signal is eliminated. The second level is structured as follows. The output signal of AOM1 (160 MHz) is split into two parts, which are separately passed through AOM2 (110 MHz) and AOM3 (110 MHz). Then, the purified optical frequency signal is output for subsequent dissemination. The output of AOM2 and 90 % of the incoming signal are sent to SMC3 for heterodyne detection via PD2, and the output signal of PD2 is amplified by filtering and sent to the phase compensation module. The phase compensation module transmits real-time demodulated phase information to AOM2. Additionally, the algorithm processes this phase data and disseminates it back to AOM3, enabling compensation for residual frequency and phase errors, as well as addressing phase cycle slips.

The frequency compensation module, as shown in Fig. 2(a), is implemented on a platform based on an Xilinx Zynq7000 FPGA. The signal is acquired by an analogue-to-digital converter (ADC). Its in-phase and quadrature (I/Q) components are extracted via multiplication with the sine and cosine of the reference signal, which has a frequency defined by a rational fraction of 48 bits chosen by the user ($f_{ref} = k/2^{48} \cdot f_{clk}$). The I/Q signals are processed via digital low-pass filters.

with a bandwidth of 1 MHz to eliminate sum frequency components. The phase error ϕ is extracted using an arctangent operation, and the relative frequency drift f is calculated by taking the difference in the successive phase errors (numerical derivative). The difference in the phase error reflects the relative frequency drift and the residual noise from the preceding link. To mitigate residual noise from preceding links, we employ a moving average method with a window of 1.6 s. In addition, we apply a threshold judgement method to eliminate phase cycle slips. We replace the current value with the moving average result from the previous cycle, enabling more accurate identification of cycle slips in the second level of the phase-locked loop. The phase error difference, which is determined via postprocessing by the algorithm, is multiplied by a fixed gain coefficient and subsequently compensated for by AOM1 through direct digital synthesizer 1 (DDS1). The frequency compensation module using a purchased commercial high-speed signal processing modules incorporates a 100 MHz clock crystal oscillator synchronized

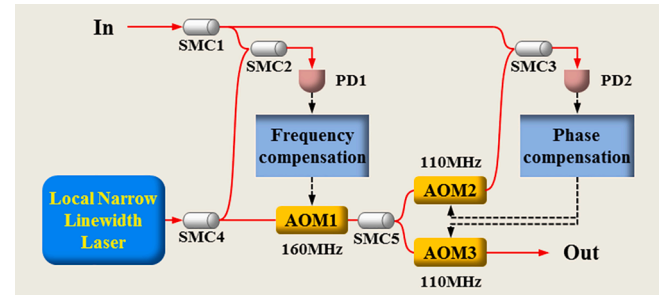


Fig. 1. Schematic diagram of the optical frequency purification process. SMC, single-mode coupler; AOM, acousto-optic modulator; PD, photoelectric detector.

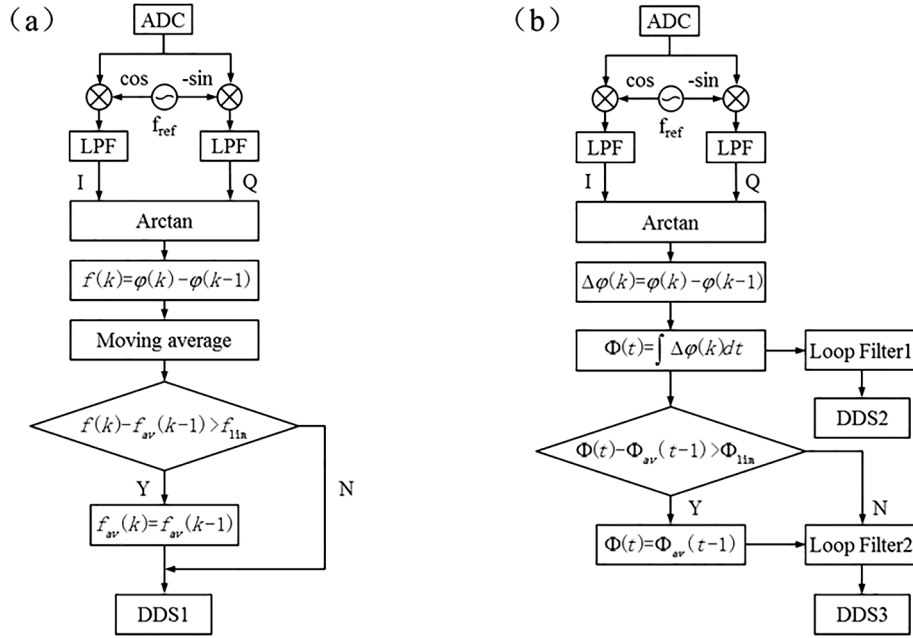


Fig. 2. Architectures and algorithmic flowcharts of the frequency compensation and phase compensation modules. (a): Frequency compensation module. (b): Phase compensation module.

with a local hydrogen clock. The sampling rate is set at 100 MHz, so applying the moving average method with a 1.6 s window involves processing 1.6×10^8 data points, which requires a 28-bit register for data storage. However, the FPGA chip has limited resources of only 16 bits. Thus, the phase error difference data are downsampled using an averaging method with 10,000 points. This downsampling strategy reduces the sampling rate to 10 kHz, and 14 bits of registers are employed to store 1.6 s of data for subsequent calculations.

The same hardware platform is used for the phase compensation and frequency compensation modules, as depicted in Fig. 2(b). Phase compensation is implemented through two symmetrical approaches in both optical paths and circuits. The signal obtained by the ADC is processed with IQ phase detection and phase unwrapping strategies [26,27]. In the first loop, real-time phase information, including cycle slips, is fully fed back through a closed loop. The data are then disseminated to AOM2 using DDS2 at a sampling rate of 1 MHz (after averaging 100 points and downsampling to 1 MHz). Simultaneously, the unwrapped phase is processed with the moving average method with a 30 ms window. If the present value surpasses the predefined threshold, the average value from the previous period is used to replace the cycle slip; however, if this value is less than the given threshold, the present value remains unchanged. Directly compensating for cycle slips with feedback within the first phase-locked loop results in reduced locks in the system, thereby diminishing system robustness. To address this issue, we utilize another open loop to replace cycle slips. The data processed by the algorithm are disseminated to DDS3 through loop filter 2, which has the same parameters as loop filter 1, and subsequently disseminated to AOM3. This approach does not affect the locks in the system and enables the substitution of cycle slips. The present system uses two DDSs to separately transmit data to the two AOMs, and system asymmetry should be eliminated by ensuring that the shortest traces are used in the optical paths and hardware circuits. The difference between the outputs of the two DDSs is approximately 4 mHz, which indicates frequency instabilities of 3.76×10^{-18} at 1 s and 1.52×10^{-20} at 20,000 s, as shown in Fig. 7. This instability (dark blue line) is less than that of the noise floor of the dissemination system (cyan line), indicating that the present hardware system does not impact the frequency instability of the dissemination process.

3. Experiments and results

We studied the purification performance of optical frequency disseminated over a 106 km communication fiber link. This fiber link, primarily buried 20 cm underground along a highway, experiences a total loss of 30 dB. The optical connectors used at the ends of the fiber are FC/APC. The link extends from the National Time Service Center (NTSC) in the Lin Tong district to the Chang'an district of Xi'an, where it loops back to the Lin Tong district. The optical signal is amplified using bidirectional erbium-doped fiber amplifiers (Bi-EDFAs) in the link. The configuration of the dissemination setup is shown in Fig. 3. All components, including those of the optical source, sender, receiver in the dissemination system, and optical frequency purification system, are centralized at the NTSC in the Lin Tong district. The optical source of the dissemination system is an ultrastable laser stabilized in an optical cavity (wavelength of 1550.12 nm and linewidth of 2 Hz). The sender and receiver form a Michelson interferometer, utilizing the acousto-optic modulator (AOM) at the sender to compensate for fiber link noise. The output signal at the receiving end is split using SMC10. One portion of the signal undergoes optical frequency purification before being disseminated to the user.

The effectiveness of the frequency compensation process is crucial in the second-level of the control loop. A Mach-Zehnder-type interferometer was used to detect the beat frequencies, as shown in Fig. 1. The relative frequency drift between the local ultrastable laser and the disseminated light was detected by PD1. The compensating signal is the output signal of DDS1 as shown in Fig. 2(a). These signals are recorded by a π -type frequency counter (K&K FXE) with a gate time of 1 s, and the measurement results are illustrated in Fig. 4. Over the 40,000 s measurement period, the relative frequency drift (blue line) is 60 kHz, and the frequency change trend of the compensated signal (red line) essentially coincides with that of the original signal. According to the measurements, the residual frequency difference (green line) is approximately 50 Hz, which considerably alleviates the challenges associated with the phase-locked control strategy in the second level.

In this paper, we investigated the characteristics of the out-loop beat note signal derived from the optical frequency dissemination and purification. To evaluate the effectiveness of the purification process, we measured and compared the phase noise power spectral densities (PSDs)

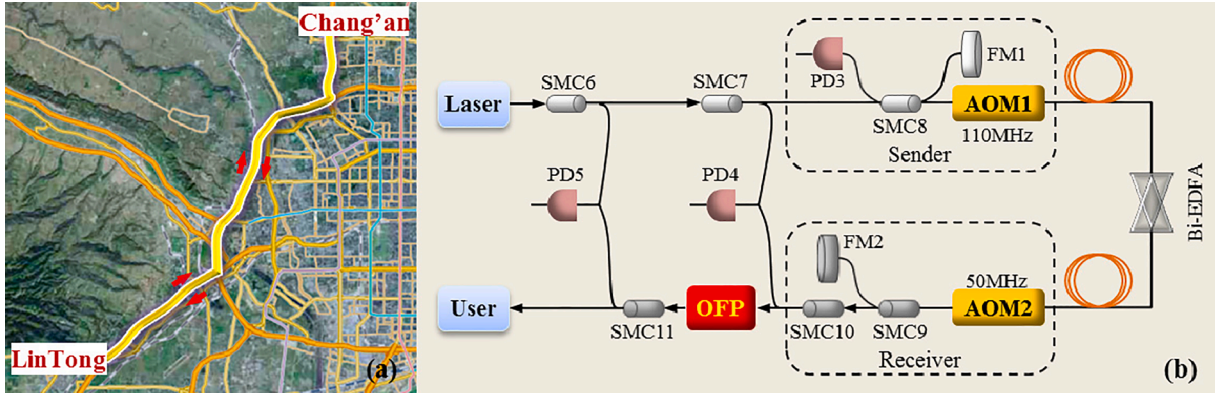


Fig. 3. (a) Map of the dissemination fiber link (based on the Baidu Map). (b) Schematic diagram of the optical frequency dissemination and purification. SMC, single-mode coupler; AOM, acousto-optic modulator; PD, photoelectric detector; FM, faraday mirror; OFF, optical frequency purification; Bi-EDFA, bidirectional erbium-doped fiber amplifier.

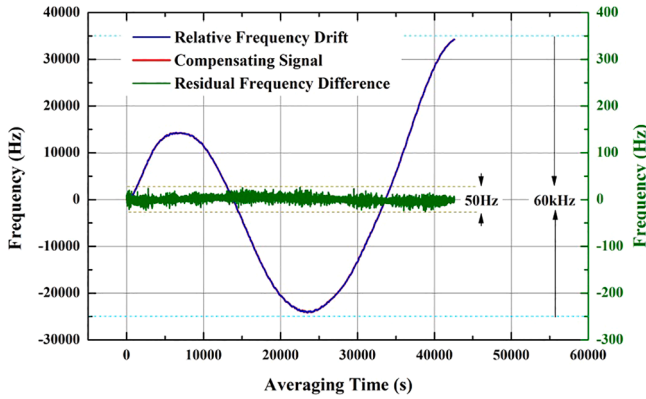


Fig. 4. Long-term frequency compensation measurements. The relative frequency drift between the local ultrastable laser and the disseminated light (blue line), the compensation signal output by the frequency compensation module (red line), and the residual frequency difference after compensation (green line).

of the local narrow linewidth laser, the out-loop signal of dissemination from the preceding link, and the purified out-loop signal. As shown in Fig. 3(b), the out-loop beat note signal of the dissemination is detected by PD4 and that of the purification is detected by PD5. Utilizing a phase

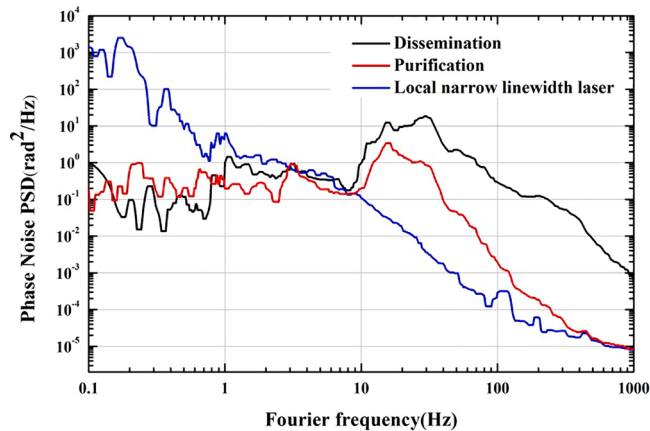


Fig. 5. Phase noise power spectral density of the optical frequency disseminated over a 106 km fiber link (black line), the purified optical frequency disseminated over a 106 km fiber link (red line), and the local narrow linewidth laser (blue line).

noise analyser (FSWP) at the local end, we captured these phase noise PSDs, as illustrated in Fig. 5. The phase noise PSD of the purification (red line) overlaps with that of the dissemination (black line) at frequencies below 10 Hz and follows the trend of the local narrow linewidth laser (blue line) at frequencies above 10 Hz. These findings indicate successful locking of the light output by the local narrow linewidth laser to the disseminated light within a 10 Hz bandwidth. In the Fourier frequency range from 10 Hz to 1 kHz, the phase noise of the purified light reduced from 20 rad to 6 rad compared to that of the disseminated light. This notable reduction in the phase noise effectively mitigates the impact of servo bump accumulation on the downstream link, resulting in an enhanced phase noise suppression ratio.

The fiber links used for long-distance optical frequency dissemination are communication fibers, which typically operate in complex environments. Optical frequency are susceptible to external human-related and environmental interference, leading to potential phase cycle slips in the dissemination system. In the operational process of the proposed experimental system, external force is manually applied by tapping or shaking the fiber to intentionally induce cycle slips, simulating the interference encountered in real-world long-distance communication fibers. The fractional frequency fluctuations of the out-loop signal from the dissemination and purification processes are monitored using a π -type frequency counter (K&K FXE) with a gate time of 1 s. The measurement results are depicted in Fig. 6(a). We evaluated the frequency fluctuations of the signals, finding that the disseminated signal (purple line) exhibits three distinct frequency peaks (P1, P2 and P3) with amplitudes of 87 Hz, 24 Hz and 202 Hz, respectively. These peaks, which exceed the typical range of frequency fluctuations expected during dissemination, are much likely caused by human interference. The fractional frequency fluctuation data are normalized and converted to phase data, as shown in Fig. 6(b). The phase fluctuations (purple line) of the disseminated signal show two cycle slips at the same locations as the frequency peaks. The algorithm in the second-level control phase compensation module is applied to remove cycle slips, and the phase fluctuations of the purified signal (dark yellow line) show that the phase was consistently maintained within the normal range, with no frequency peaks observed in the fractional frequency fluctuations (dark yellow line) as shown in Fig. 6(a).

The instabilities of the disseminated and purified optical frequency over the 106 km fiber link are depicted in Fig. 7. The Allan deviation (ADEV) should be proportional to $\frac{1}{\tau}$ for a signal dominated by phase-modulated white noise; due to the loss of phase continuity in the frequency information, the ADEV is proportional to $\frac{1}{\sqrt{\tau}}$ with increasing averaging time. The fractional frequency instability of the disseminated optical frequency (purple line) is 1.20×10^{-14} at an averaging time of 1 s, and the long-term instability influenced by cycle slips is 2.35×10^{-17}

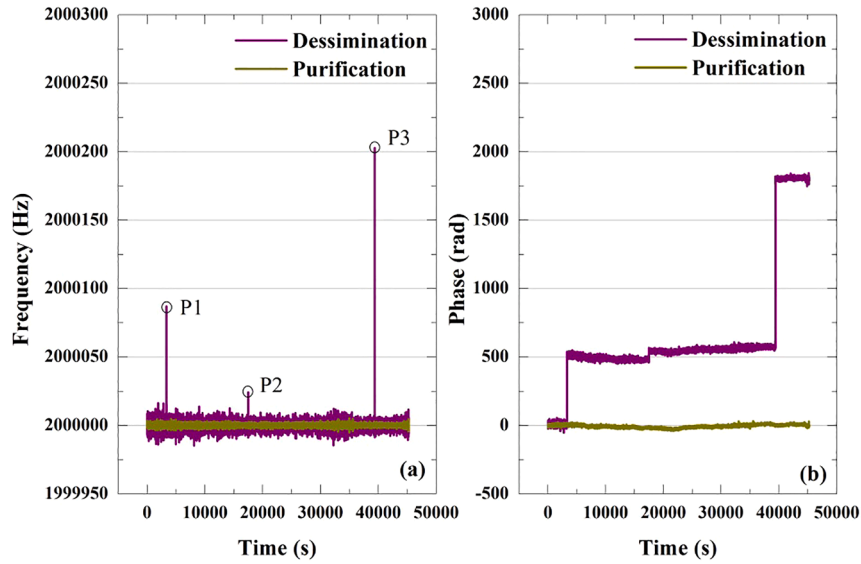


Fig. 6. (a) The fractional frequency fluctuations of an optical frequency disseminated over a 106 km fiber link (purple line) and the purified optical frequency (dark-yellow line); (b) the corresponding phase fluctuations of the disseminated (purple line) and purified optical frequencies (dark-yellow line).

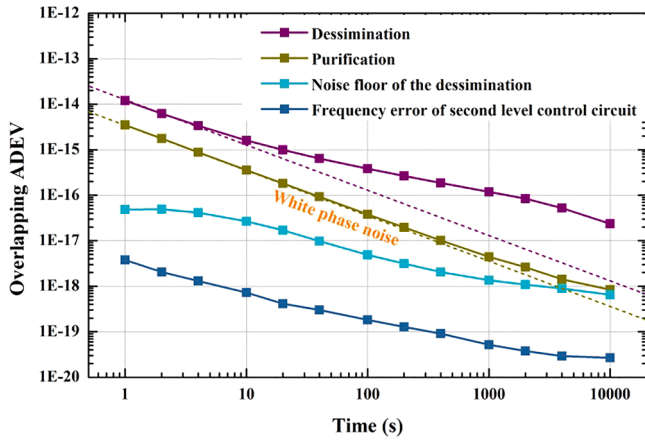


Fig. 7. Overlapping ADEV of the disseminated and purified optical frequency over the 106 km fiber link. Instabilities of the disseminated optical frequency (purple line), purified optical frequency (dark-yellow line), noise floor of the disseminated optical frequency (cyan line), and frequency error of the second-level control circuit (dark-blue line).

at an averaging time of 10000 s. As the phase noise suppression ratio increases after the disseminated optical frequency is purified, the overlapping ADEV of the purified optical frequency (dark yellow line) is 3.50×10^{-15} at an averaging time of 1 s, which is 3.4 times greater than that of the disseminated optical frequency, and the overlapping ADEV is 8.43×10^{-19} at an averaging time of 10,000 s when the phase is continuous. This exceptional instability demonstrates that the optical frequency can maintain coherence even when it is disseminated over a 106 km fiber link. For further evaluation of this compensation system, we also evaluate the accuracy of optical frequency transfer [28]. Based on the fractional frequency fluctuation data presented in Fig. 6(a), we calculated the standard deviation of the disseminated optical frequency signal to be 0.98 mHz, with a corresponding accuracy of 5.07×10^{-18} . After correcting for cycle slips, the standard deviation of the purified optical frequency signal was reduced to 0.39 mHz, achieving an accuracy of 2.02×10^{-19} . This represents a 2.5-fold improvement in accuracy compared to the disseminated optical frequency signal.

4. Conclusion and perspectives

We studied an optical frequency disseminated over a 106 km communication fiber link to evaluate the performance of the proposed optical frequency purification system. The optical frequency purification process, as detailed in our investigation, locks the phase of the local narrow linewidth laser to that of the prior disseminated optical frequency within a bandwidth of 10 Hz and provides an optical frequency with less phase noise for the subsequent link. Moreover, the proposed system eradicates cycle slips in the system induced by fiber link interference, thereby preserving the coherence of the optical frequency. We also determined the overlapping ADEV of the fractional frequency in the data analysis presented. The instability of the optical frequency disseminated over the 106 km fiber link is 1.57×10^{-14} with a 1 s averaging time and scales as $\frac{1}{\sqrt{\tau}}$. Following optical frequency purification, the frequency instability improves to 6.0957×10^{-15} with a 1 s averaging time, scaling as $\frac{1}{\tau}$ to 6.15×10^{-19} at a 10,000 s averaging time. These results demonstrate that the proposed system can provide more accurate optical frequency disseminated over long-distance fiber links. In real-world deployment scenarios, our project will include the deployment of time-frequency transfer equipment to transmit a synchronized 10 MHz signal from the hydrogen clock to the machine room where the devices are located. This signal will serve as the synchronization source for the 100 MHz clock crystal oscillator. To enhance the long-term instability of both the disseminated and purified signals, aiming for an instability on the order of 10^{-21} , which is typical of transfer system noise, the impact of temperature changes on the asymmetrical components of the optical modules must be mitigated. Additionally, given the diverse nature of external interferences in optical frequency dissemination systems composed of fiber links, the de-hopping method should be adjusted and optimized to effectively address various types of cycle slips.

CRedit authorship contribution statement

Dan Wang: Writing – original draft, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Jie Liu:** Investigation. **Dongdong Jiao:** Investigation. **Qi Zang:** Data curation. **Xiang Zhang:** Investigation. **Qian Zhou:** Data curation. **Yucan Zhang:** Software. **Guanjun Xu:** Investigation. **Ruifang Dong:** Supervision. **Tao Liu:** Supervision, Funding acquisition, Conceptualization. **Shougang Zhang:**

Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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